Reinforcement Learning Summative Assignment Report

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Video Recording: https://youtu.be/DdA-wsaHpEo
GitHub Repository: https://github.com/thepatrickniyo/patrick-niyogitare_rl_summative

1. Project Overview

This project implements an Adaptive **AI Learning & Mentorship Optimization system** aligned with **CodetyAI**: a simulated single-student journey through lessons, projects, mentor sessions, and revision. The goal is to maximize employability outcomes (skill, confidence, completed projects) for STEM learners—a proxy for reducing youth unemployment through better sequencing of learning interventions. We trained and compared **DQN** (value-based), **REINFORCE** (Monte Carlo policy gradient), **PPO**, and **A2C** on the same Gymnasium environment using Stable-Baselines3 (and a custom REINFORCE trainer), with ten hyperparameter runs per algorithm logged to results/training_log.csv.

2. Environment Description

The CodetyAI environment (**CodetyAILearningEnv**) simulates one learner with stochastic transitions: the agent observes normalized skill, engagement, project count, mentorship count, confidence, and program progress, and receives sparse rewards for projects, skill jumps, job-readiness, and penalties for engagement loss and dropout.

a. Agent(s)

The agent is a discrete policy over five platform actions. It does not model human mentors directly; it learns which intervention to schedule next given the current student state, with the objective of maximizing cumulative reward (aligned with job-readiness).

b. Action Space

Desecrete, 5 Actions

ID	Action
0	Beginner lesson
1	Advanced lesson
2	Hands-on project
3	Connect mentor
4	Recommend revision

c. Observation Space

Box(6,), values in **[0, 1]**: skill/100, engagement/2, projects/cap, mentorship/cap, confidence/100, step progress (step / max episode length). The agent does not observe the full latent "world"—only these summaries—so partial observability is limited to what is encoded in the vector.

d. Reward Structure

+10 when a hands-on project is completed successfully in the transition model.

+15 when skill increases by ≥ 3 in one step.

+20 when the student becomes job-ready (skill ≥ threshold, confidence ≥ threshold, minimum projects).

−10 when engagement drops one level.

−20 on dropout (terminal failure).

3. System Analysis And Design

a. Deep Q-Network (DQN)

DQN uses an **MLP policy** via SB3, **experience replay**, **target network** updates, and **ε-greedy exploration** with linear decay (exploration_fraction, exploration_final_eps). Training minimizes TD error on minibatches sampled from the replay buffer.

b. Policy Gradient Method ([REINFORCE/PPO])

Custom **Monte Carlo REINFORCE** with Adam, optional entropy bonus, gradient clipping, and an MLP policy (two hidden layers + Tanh). Episodes: **300** per run for the sweep below; evaluation **20** episodes.

4. Implementation

a. DQN

Learning Rate	Gamma	Replay Buffer	Batch Size	Exploration Strategy (ε schedule)	Mean Reward

1e-4	0.99	50,000	32	frac 0.20 → ε_end 0.05	201.75
3e-4	0.99	100,000	64	frac 0.15 → ε_end 0.02	201.75
5e-4	0.995	80,000	128	frac 0.25 → ε_end 0.08	238.25
2.5e-4	0.99	200,000	256	frac 0.10 → ε_end 0.01	201.75
1e-3	0.97	40,000	32	frac 0.35 → ε_end 0.10	247.75
6e-5	0.999	150,000	64	frac 0.18 → ε_end 0.04	199.50
4e-4	0.98	60,000	48	frac 0.22 → ε_end 0.06	231.25
2e-4	0.99	120,000	96	frac 0.12 → ε_end 0.03	201.75
7e-4	0.985	90,000	128	frac 0.28 → ε_end 0.07	247.75
1.5e-4	0.993	250,000	64	frac 0.08 → ε_end 0.015	202.00

b. REINFORCE

Learning Rate	Gamma	Hidden Units	Entropy Coef.	Max Grad Norm	Mean Reward
1e-3	0.99	128	0.0	1.0	218.25
3e-4	0.995	256	0.01	0.5	201.75
5e-4	0.98	64	0.005	2.0	201.75
1e-4	0.999	192	0.0	1.5	156.00
7e-4	0.99	160	0.02	0.3	201.75
2e-3	0.97	96	0.015	0.8	206.25
4.5e-4	0.992	200	0.003	1.2	201.75
8e-5	0.996	128	0.0	2.5	201.75

6e-4	0.988	144	0.008	0.65	201.75
1.5e-3	**0.985**	**112**	**0.025**	**0.45**	**234.75**

c. PPO

Learning Rate	N Steps	Batch Size	Clip Range	Entropy Coef.	Mean Reward
3e-4	512	64	0.20	0.0	201.75
1e-4	1024	128	0.15	0.005	201.75
5e-4	256	32	0.25	0.01	218.25
2.5e-4	2048	256	0.10	0.0	201.75
8e-4	128	16	0.30	0.02	201.75
4e-4	768	96	0.18	0.002	218.25

6e-5	1536	192	0.12	0.0	201.75
1e-3	384	48	0.22	0.015	201.75
2e-4	640	80	0.16	0.008	218.25
3.5e-4	896	112	0.19	0.003	201.75

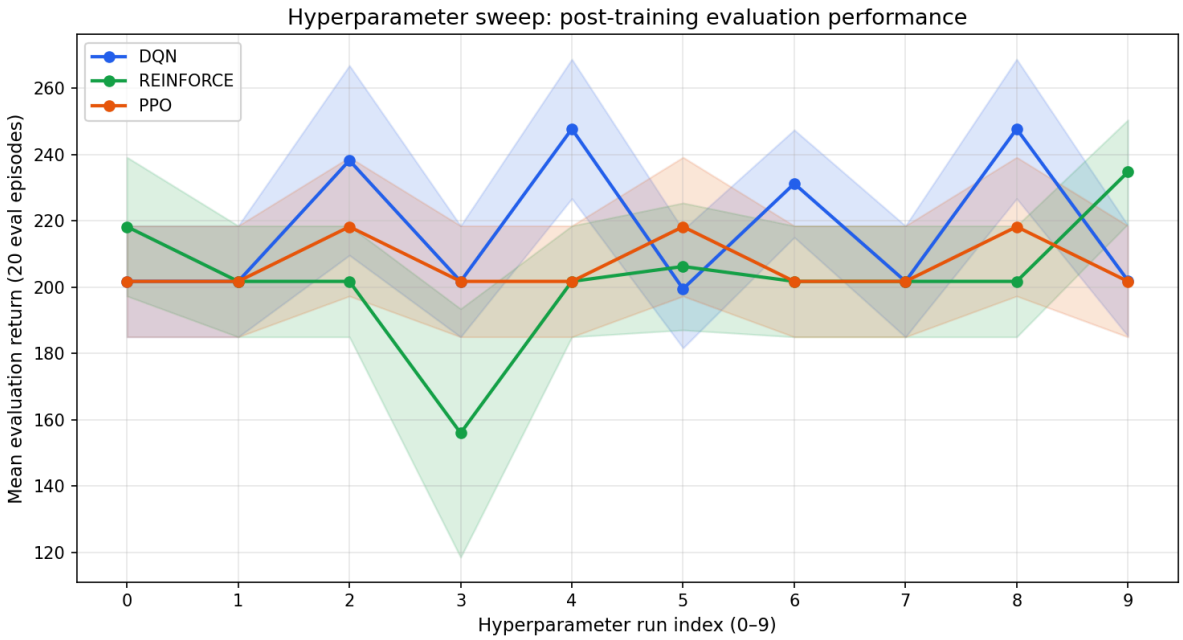
5. Results Discussion

Describe and interpret every visualization

a. Cumulative Rewards

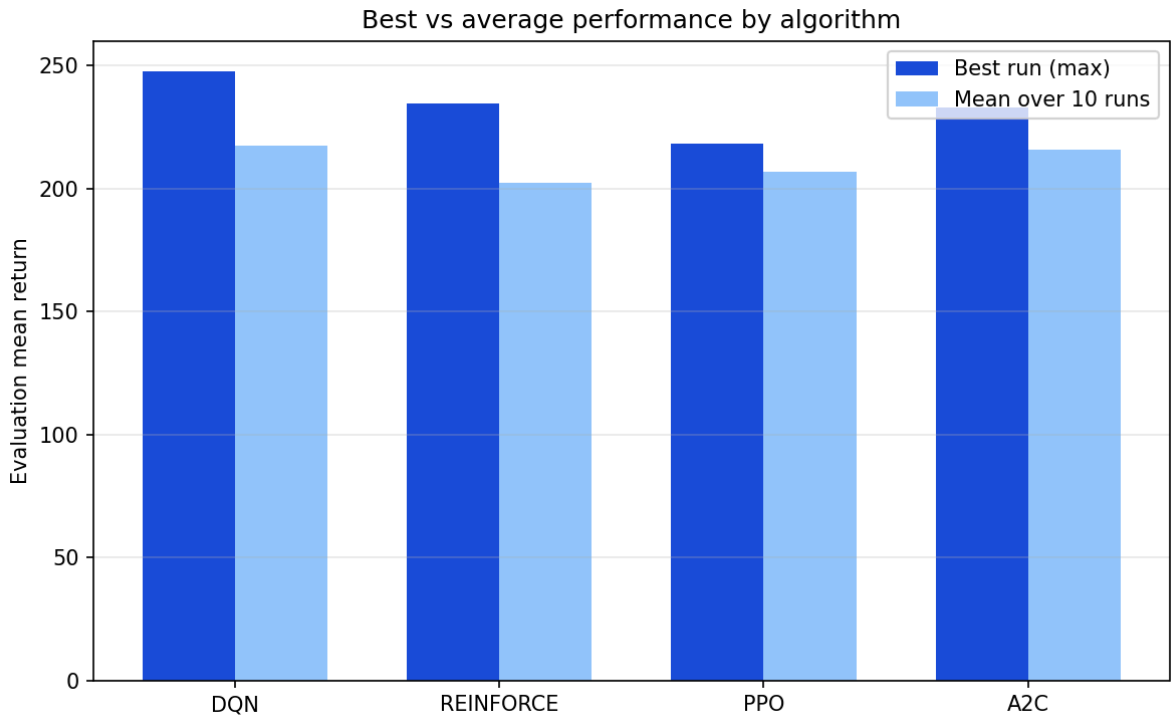
a.1 Evaluation return across hyperparameter runs

The line plot below shows **post-training evaluation mean return** (20 episodes each) for each of the **10 hyperparameter rows** per algorithm. Shaded bands use **±1 std** from the evaluation rollout reported in the log. This summarizes **how sensitive each method is to the chosen hyperparameter grid**, not the online learning curve during backprop (for that, use TensorBoard logs under results/tb/ if enabled during training).



Interpretation: DQN (blue) reaches the highest peaks (runs 4 and 8). REINFORCE (green) is more variable—run 3 underperforms and run 9 peaks. PPO (orange) is relatively flat around ~201–218 for most runs, with stronger runs at indices 2, 5, and 8. The bands show that **evaluation noise** is moderate; several DQN/PPO configurations cluster near the same mean when the policy quality is similar.

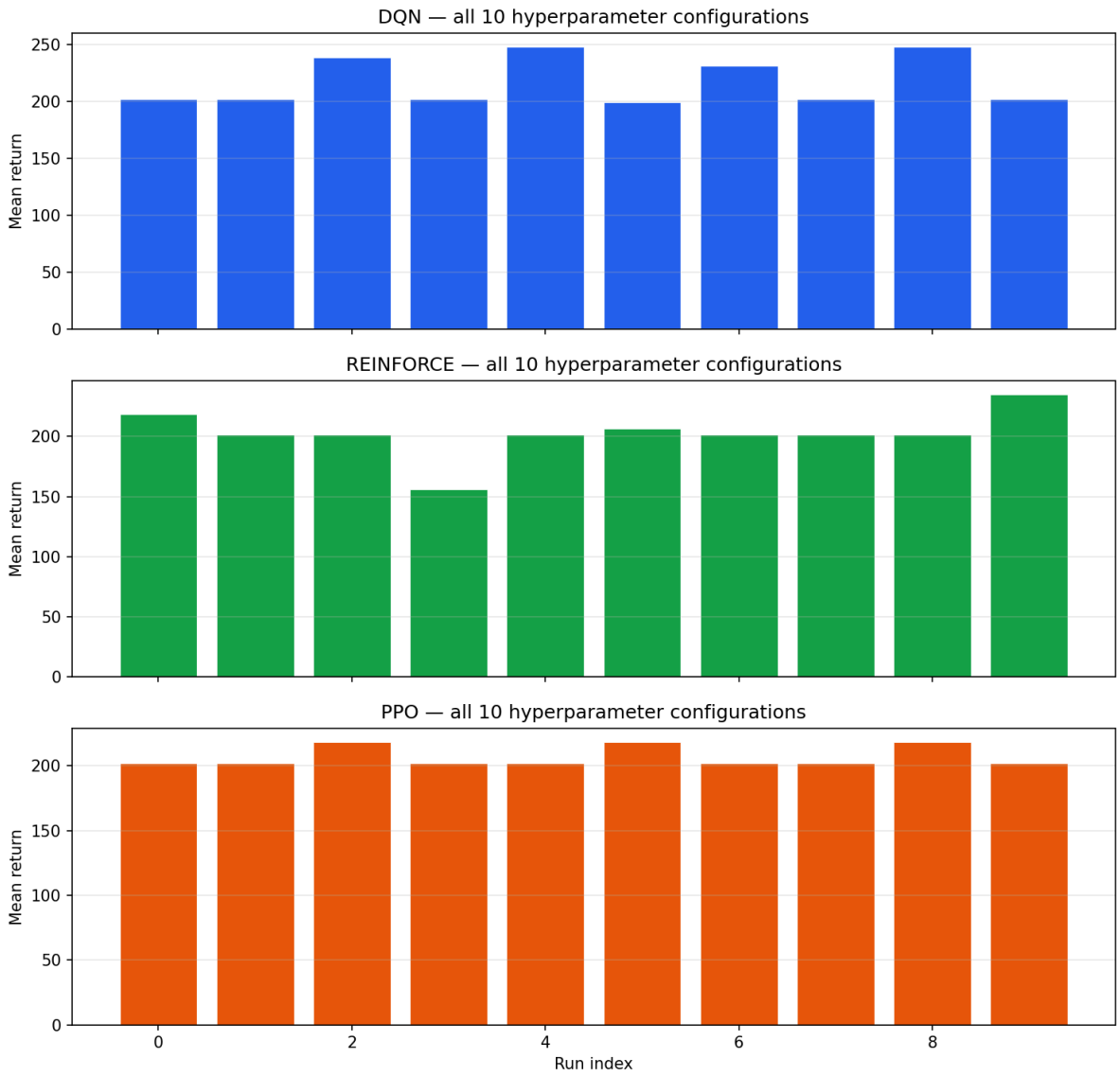
a.2. **Best vs mean return by algorithm**



Interpretation: For each algorithm we compare the **best** of the 10 runs to the **mean** over all 10. A large gap

suggests **high variance** across hyperparameters; a small gap suggests **robust** settings. A2C is included from the same log (policy-gradient family alongside PPO).

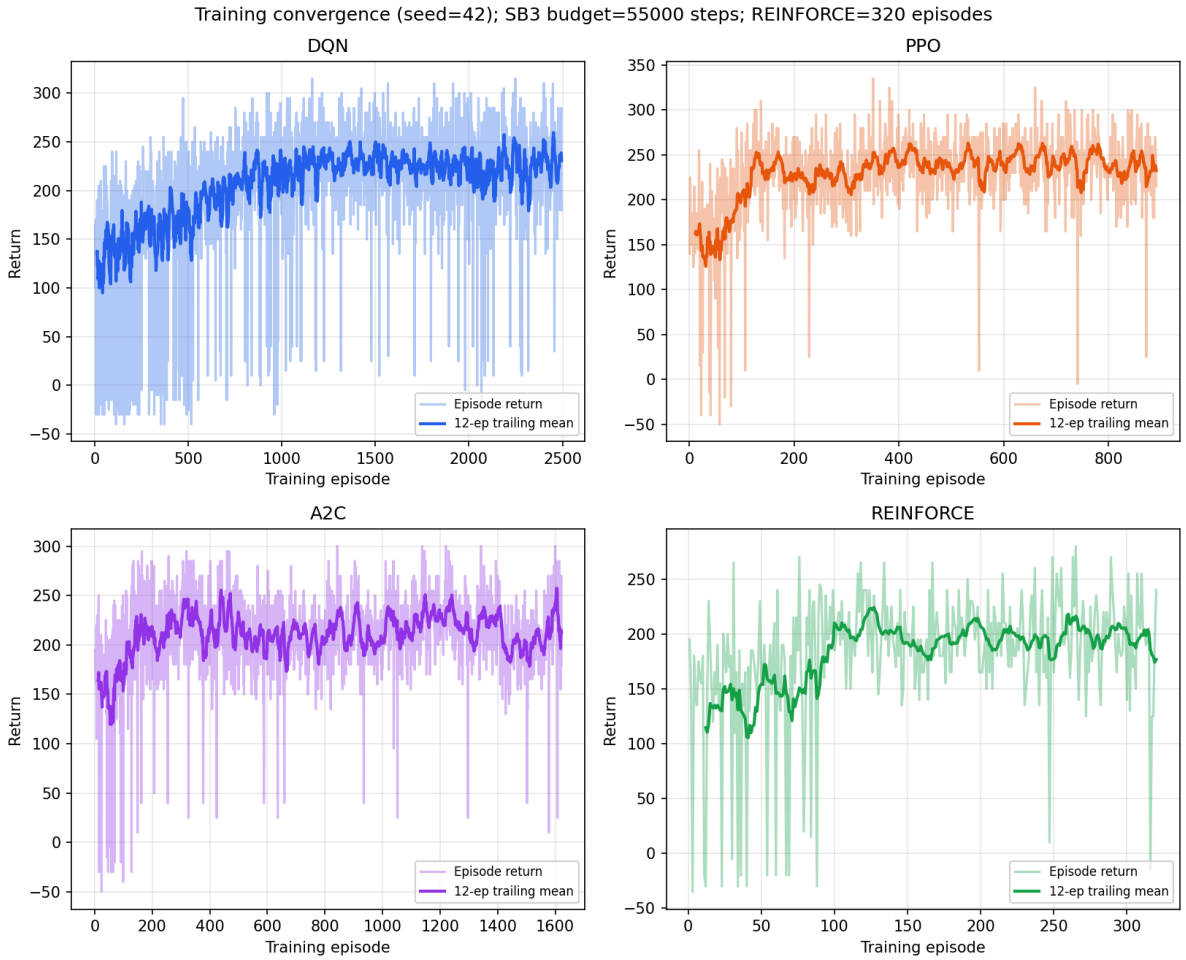
a.3 Per-run bar charts (DQN, REINFORCE, PPO)



Interpretation: Side-by-side bars make it easy to spot **outlier runs** (e.g. REINFORCE run 3) and **duplicate peaks** (DQN runs 4 and 8). This complements the line plot for slide/report layouts.

b. Episodes To Converge

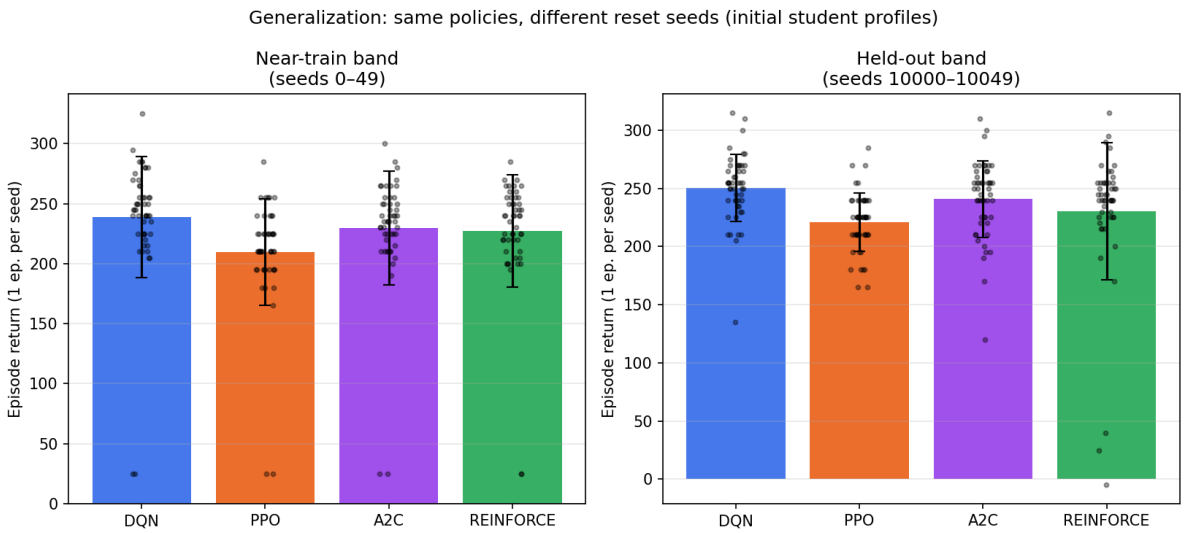
We logged **per-episode return** during a controlled rerun



with **shared random seed (42)** and **matched compute for SB3**: **55 000 environment steps** each for **DQN** (hyperparameter row 4), **PPO** (row 2), and **A2C** (row 2). **REINFORCE** (row 9) ran for **320 weight updates** (one full episode per update). The figure below shows **raw episode return** (faint) and a **12-episode trailing mean** (bold) on each subplot.

Under the **same SB3 step budget**, **PPO's smoothed curve reaches a high plateau quickly** (90% of its peak trailing mean by episode **124** of logged completions). **REINFORCE** hits a similar threshold by episode **100** but uses **episodes as the optimisation unit** (320 updates total), with **higher tail variance** (CV 0.23). **A2C** is slower to reach the 90%-of-peak-smoothed level (episode **259**). **DQN** logs **many short episodes** within 55k steps (early termination is common), so its **episode index** is **not directly comparable** to length-biased PPO/A2C counts; its trailing mean still trends upward, with a **noisier tail** (CV 0.22). For fair comparison across value-based and on-policy methods, compare **environment steps**; for **REINFORCE**, compare **policy-gradient updates** (episodes).

c. Generalization



No algorithm **collapses** on the far seed band; mean returns are **similar or slightly higher** there for these checkpoints, so **catastrophic sensitivity to this seed offset** is not observed. **REINFORCE** shows the **smallest** absolute gap between bands (~3 return units, ~1.4%), while DQN/PPO/A2C shift by ~11 points (~5%). **Lower far-band std for DQN and PPO** suggests **less dispersion** across those seeds than in 0–49 for those runs. These results **do not** substitute for evaluation on **held-out initial conditions**; they complement the fixed-seed training curves in **(e)** by showing **cross-seed spread** at deployment time.

6. Conclusion and Discussion

Across the logged sweeps on this environment, **DQN achieved the highest evaluation mean return** (up to **247.75**), followed by **REINFORCE** (best **234.75**) and **PPO** (best **218.25**). Value-based DQN benefited from stable replay and ε-decay on this discrete, moderate-dimensional task; REINFORCE showed higher variance and sensitivity to learning rate and entropy regularization; PPO remained stable but did not exceed the best DQN run under the same timesteps and evaluation protocol.

Strengths: DQN sample efficiency via replay; PPO stable updates; REINFORCE simple and interpretable.
Weaknesses: REINFORCE variance; all methods are sensitive to reward scale and episode length; simulation is not real student data.

Improvements: longer training, curriculum on student profiles, offline RL from logs, and human-in-the-loop constraints for deployment.